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# OS

# MINI-PROJECT READ ME

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# project overview

This project is an automated Linux-based auditing solution designed to collect hardware and software information, generate formatted reports, and support remote monitoring and scheduled execution. It is developed using modular shell scripting to assist system administrators and cybersecurity analysts in maintaining asset inventories and performing risk assessments.

# features

Modular Audit Scripts: Separate modules for Hardware and Software auditing with both "Short" (summary) and "Full" (detailed) reporting options

Interactive Menus: Two management interfaces—a CLI-based terminal menu (main.sh) and a graphical dialog-based menu (gui.sh)

Automated Monitoring: Scheduled execution via cron jobs with integrated email reporting.

Professional Reporting: Reports are generated with headers, timestamps, and colorized output, then saved automatically to local logs.

Security & Integrity: Includes root privilege checks, error handling with traps, and log rotation for system persistence.

# Installation & Prerequisites

## 1. System Requirements

OS: Linux (Ubuntu, Kali, or similar Debian-based distributions).
Privileges: The scripts must be run with root/sudo privileges to access system hardware and log files.

## 2. Required Packages

Ensure the following standard tools are installed:
sudo apt update
sudo apt install mailutils msmtp dialog lshw dmidecode

## 3. Setup

Extract the project files and grant execution permissions to all scripts:
chmod +x *.sh

# Configuration

## Email Reporting

The scripts use mailx or msmtp to send reports.

- Manual Export: Upon completing a manual audit, the script will prompt if you wish to email the result.
- Automated Export: The automation.sh script is pre-configured to send reports to our email, You can modify the RECIPIENT variable in that file to change the target address.

## Log Files

Audit data and errors are stored in the following locations:

- Software Logs: /var/log/software_audit.log
- Hardware Logs: /var/log/system_report.log
- Report Outputs: Temporary report files are stored in /tmp/ (e.g., short_SOFTWARE_Format.txt).

# Configuration



```
defaults
auth            on
tls             on
tls_trust_file /etc/ssl/certs/ca-certificates.

account gmail
host smtp.gmail.com
port 587
from your@email.com
user your@email.com
password yourpassword

account default : gmail
```

# How to Run

Option 1: Interactive Master Control (Recommended)

Launch the main management menu to select specific audit tasks:

sudo ./main.sh

Option 2: Graphical User Interface

For a more visual experience using the dialog tool:

sudo ./gui.sh

Option 3: Individual Modules

You can run any audit module directly:

- sudo ./softwareshort.sh (Concise software audit)
- sudo ./softwarefull.sh (Detailed software inventory)
- sudo ./hardshort.sh (Concise hardware summary)
- sudo ./hardlong.sh (Full technical hardware specs

# 1. Host and Guest Network Synchronization

To ensure both machines can communicate, the VirtualBox network settings were adjusted:

- Shutdown the VM: The machine must be powered off to change network adapters.
- Switch to Bridged Adapter: Changed the network mode from NAT to Accès par pont (Bridged Adapter).
- Enable Connectivity: Ensured "Virtual Cable Connected" was checked to allow the VM to receive an IP from the local router.

# Installing and Enabling SSH Server (On Guest VM)

Since the ssh.service was missing, the OpenSSH server was installed to listen for incoming connections:

```
# Update package lists
sudo apt update

# Install OpenSSH Server
sudo apt install openssh-server -y

# Enable and start the service
sudo systemctl enable --now ssh

# Verify service status
sudo systemctl status ssh
```

# 3. Identifying the Remote IP Address:

To find the internal IP assigned to the VM, the following command was used:

```
hostname -I
```

# 4. Setting Up SSH Key-Based Authentication (Passwordless Login):

To allow scripts to run without manual password entry, an SSH key pair was established:

### Generate RSA Keys (On Host Machine)

```
ssh-keygen -t rsa -b 4096
# Pressed Enter for all prompts to use default paths and no passphrase.
```

## Transfer Public Key to Guest VM:

```
ssh-copy-id sandy@<GUEST_IP_ADDRESS>
```

## Verification:

```
ssh sandy@<GUEST_IP_ADDRESS>
```

## result:

```
pc@DESKTOP-5RMSR7D C:\Users\pc>
```

## sendremote script :

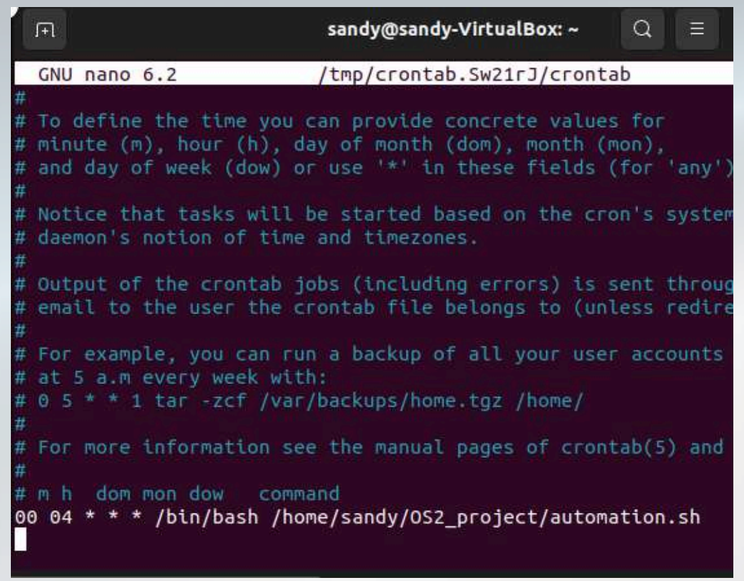
```
sandy@sandy-VirtualBox:~/OS2_project$ ./sendremote.sh
NSCS Remote Transfer
Enter Target IP Address: 10.195.95.235
Enter Remote Username: pc
Target Port: 22
Is the target Windows? (y/n): y
Enter Windows Path (e.g: C:\Users\Public): C:\Users\pc

Connecting to 10.195.95.235 on port 22...
pc@10.195.95.235's password:
Permission denied, please try again.
pc@10.195.95.235's password:
long_software_Format.txt                    100% 6362   361.0KB/s   00:00
Sent: long_software_Format.txt
pc@10.195.95.235's password:
Long_Hardware_Format.txt                    100%   12KB 712.0KB/s   00:00
Sent: Long_Hardware_Format.txt
pc@10.195.95.235's password:
short_SOFTWARE_Format.txt                   100% 1582   135.4KB/s   00:00
Sent: short_SOFTWARE_Format.txt
pc@10.195.95.235's password:
Short_Hardware_Format.txt                   100% 3774   246.0KB/s   00:00
Sent: Short_Hardware_Format.txt
```

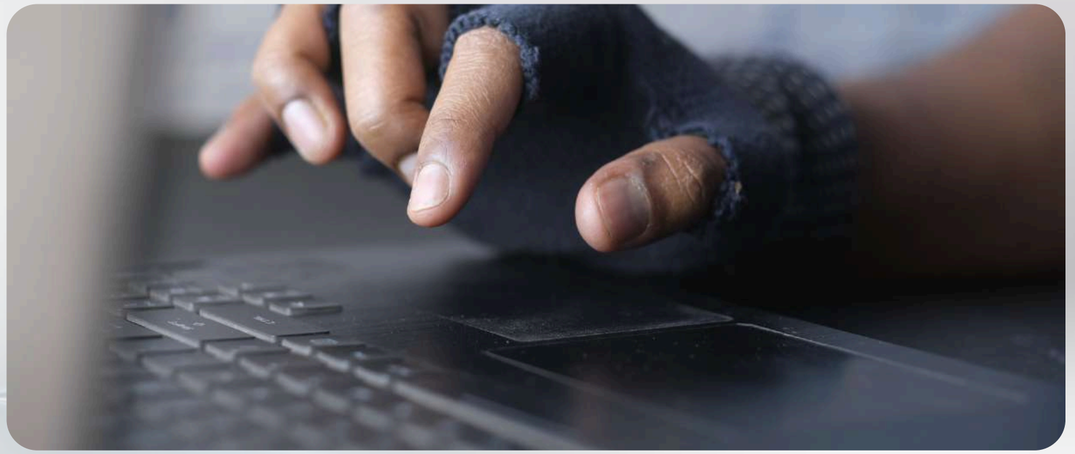
# Automation (Cron Configuration)

To automate the system audit (e.g., daily at 04:00 AM), add the automation.sh script to your crontab:

1. Open the crontab editor: sudo crontab -e
2. Add the following line:

00 04 * * * /absolute/path/to/your/project/automation.sh

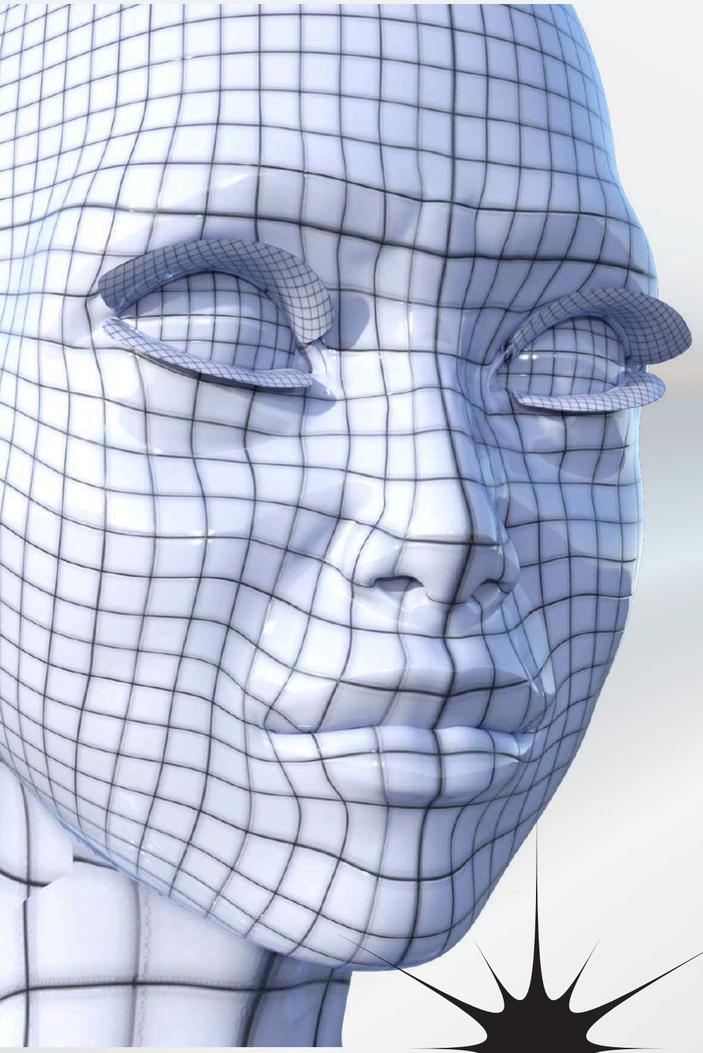


# Project Structure

- main.sh: CLI-based Master Control menu.
- gui.sh: Graphical dialog menu interface.
- automation.sh: Automation script for cron and mass email reporting.
- softwareshort.sh / softwarefull.sh: OS and package auditing modules.
- hardshort.sh / hardlong.sh: Hardware and peripheral auditing modules.

° sendremote.sh:used to send report files to another machine in the same network using ssh and scp



# Thank You

SHELL SCRIPTING